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Assignment 1: Exploring AWS Global Infrastructure

Overview

In this assignment, you will take your first steps into cloud computing by exploring pre-created IAM Users and Groups, Inspecting IAM policies as applied to the pre-created groups, following a real-world scenario, adding users to groups with specific capabilities enabled, Locating and using the IAM sign-in URL, Experimenting with the effects of policies on service access, deploying a simple static website with Amazon S3, and launching your first virtual machine (EC2 instance). By the end of the assignment, you will have hands-on experience with core cloud services and an understanding of what makes cloud computing different from running applications on your personal computer.

Deliverables

Each student (or group) must submit the following:

1. **Presentation Slides (8–12 slides maximum)**
    - Title slide with student name(s), course, and assignment title.
    - Clear explanation of each completed task (AWS Console exploration, add User to group, S3 website deployment, EC2 instance setup, termination or stop steps, and reflection).
    - Screenshots or videos of key steps/results (e.g., Add user to group, AWS Console, S3 bucket, website endpoint, EC2 stop or terminal output).
    - Reflection question responses (may be included as slides or an appendix).
    - Lessons learned and challenges encountered.
2. **Demonstration (in-class or recorded)**
    - A 5–8-minute walkthrough of the slides and discussion of your results.
    - Optional short demo (if possible) showing the live static website or EC2 instance connection.

Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (5–8 minutes).
- Be prepared to answer questions about the process and challenges.

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Detailed Grading Rubric (100 points total)

All team members are required to contribute to the presentation. The final score will be proportionally adjusted based on the number of members who present. For example, if a group consists of five students but only four deliver the presentation, and the maximum score is 100 points, the adjusted score will be calculated as:

$$Adjusted\ Score\ =\ (\frac{4}{5})\ \times 100\ =\ 80$$

1. Task Completion (80 points total)

Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots, output).

- Task 1 – AWS Account Setup & Console Exploration
  - AWS account created / logged
- Task 2 -- In the search box to the right of Services, search for and choose IAM to open the IAM console. In the navigation pane on the left, choose Users. The following IAM Users have been created for you: **user-1, user-2, user-3.**
  - Choose the **user-1** link: This will bring to a summary page for user-1. The Permissions tab will be displayed. Notice that user-1 does not have any permissions.
  - Choose the **Groups** tab: user-1 also is not a member of any groups.
  - Choose the **Security credentials** tab: user-1 is assigned a **Console password**.
  - In the navigation pane on the left, choose **User groups**: The following groups have already been created for you: EC2-Admin, EC2-Support, S3-Support
  - Choose the **EC2-Support** group link: This will bring you to the summary page for the **EC2-Support** group.
  - Choose the **Permissions** tab: This group has a Managed Policy associated with it, called **AmazonEC2ReadOnlyAccess**. Managed Policies are pre-built policies (built either by AWS or by your administrators) that can be attached to IAM Users and Groups. When the policy is updated, the changes to the policy are immediately apply against all Users and Groups that are attached to the policy.
  - Choose the plus (+) icon next to the AmazonEC2ReadOnlyAccess policy to view the policy details. **Note**: A policy defines what actions are allowed or denied for specific AWS resources. This policy is granting permission to List and Describe information about EC2, Elastic Load Balancing, CloudWatch and Auto Scaling. This ability to view resources, but not modify them, is ideal for assigning to a Support role. The basic structure of the statements in an IAM Policy is: **Effect** says whether to *Allow* or *Deny* the permissions. **Action** specifies the API calls that can be made against an AWS Service (eg *cloudwatch:ListMetrics*). **Resource** defines the scope of entities covered by the policy rule (eg a specific Amazon S3 bucket or Amazon EC2 instance, or * which means *any resource*).
  - Choose the minus icon (**-**) to hide the policy details.
  - In the navigation pane on the left, choose User groups:

- Choose the S3-Support group link and then choose the Permissions tab. The S3-Support group has the AmazonS3ReadOnlyAccess policy attached.
- Choose the plus (+) icon to view the policy details. This policy grants permissions to Get and List resources in Amazon S3.
- Choose the minus icon (-) to hide the policy details.
- In the navigation pane on the left, choose **User groups**:
- Choose the **EC2-Admin** group link and then choose the **Permissions** tab: This Group is slightly different from the other two. Instead of a *Managed Policy*, it has an **Inline Policy**, which is a policy assigned to just one User or Group. Inline Policies are typically used to apply permissions for one-off situations.
- Choose the plus (+) icon to view the policy details: This policy grants permission to view (Describe) information about Amazon EC2 and also the ability to Start and Stop instances.
- Choose the minus icon (-) to hide the policy details.

- Task 3 - Task 2: Add Users to Groups
  - You have recently hired user-1 into a role where they will provide support for Amazon S3. You will add them to the S3-Support group so that they inherit the necessary permissions via the attached *AmazonS3ReadOnlyAccess* policy.
  - Ignore any "not authorized" errors that appear during this task. They are caused by your lab account having limited permissions and will not impact your ability to complete the lab.
  - **Add user-1 to the S3-Support Group**: In the left navigation pane, choose User groups. Choose the **S3-Support** group link. Choose the Users tab. In the Users tab, choose Add users.
  - In the Add Users to S3-Support window, configure the following: Select user-1. At the bottom of the screen, choose Add users. In the Users tab you will see that user-1 has been added to the group.
  - **Add user-2 to the EC2-Support Group**: You have hired user-2 into a role where they will provide support for Amazon EC2. Using similar steps to the ones above, add **user-2** to the **EC2-Support** group. user-2 should now be part of the **EC2-Support** group.
  - **Add user-3 to the EC2-Admin Group:** You have hired user-3 as your Amazon EC2 administrator, who manage your EC2 instances. Using similar steps to the ones above, add user-3 to the **EC2-Admin** group. user-3 should now be part of the **EC2-Admin** group.
  - In the navigation pane on the left, choose **User groups**. Each Group should now have a **1** in the Users column, indicating the number of Users in each Group.
- Task 4 – Static Website with S3
  - Click on existing S3 bucket to utilize the resources successfully.
  - Static website hosting enabled correctly
  - index.html uploaded and accessible
  - Website endpoint tested and working
- Task 5 – Launching Existing EC2 Instance
  - Test with Launching new Instance by correctly (AMI, type EC2 group)

- Evidence of system info including error or warning if new EC2 not launched successfully.
- Task 6 – Terminating or Stop Existing EC2 Instance
  - Proper termination or Stop EC2 using Console
  - Screenshot or log confirming termination
- Task 7 – Reflection Questions
  - Complete and thoughtful answer for question 1
  - Complete and thoughtful answer for question 2

2. Presentation Quality (10 points total)

- Clear structure (intro, results, discussion, conclusion)
- Slides visually organized and professional, for example, adding pictures, transitions, animations
- Screenshots or videos included as evidence
- Proper timing (within 5–8 min)

3. Professionalism & Delivery (10)

- Confident and clear presentation delivery
- Ability to answer questions from peers/instructor
- Team collaboration and equal participation

Submission Instructions

- Submit slides (PowerPoint or PDF) to the LMS (Moodle) by the deadline.
- Each group will present during the next class session

Task 1-3: AWS Account Setup & Console Exploration

Become familiar with the AWS environment.
Instructions:
1. Log in to your AWS account (or create one).
2. Navigate to the AWS Management Console.
3. Use the search bar at the top to find different services.
4. Pick at least two services to explore. Suggested ones include:
   - EC2 (Elastic Compute Cloud)
   - S3 (Simple Storage Service)
   - IAM (Identity and Access Management)
5. For each service, click into it and read the description on the overview page.

Task 4: Deploy a Static Website with Amazon S3

Learn how to host content in the cloud.
Instructions:
1. In the AWS Console, search for **S3** and open the service.
2. Click on any one existing bucket to open the bucket.

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4. Upload a simple **index.html** file (you can use and modify the provided HTML file).
5. Enable static website hosting in the bucket properties.
6. Click on Open link (top right side) button to test in your browser.

Task 5: Launching EC2 Instances

Objective: Understand and analyze cloud-based virtual machines launching steps.

1. AWS Management Console (GUI)

This is the most common and straightforward method. It's great for beginners and for tasks where you need a visual interface.

Instructions:

1. Navigate to the **EC2 service** in the AWS Console.
2. Click **Launch Instance**.
3. Choose an **AMI (Amazon Machine Image)**, such as **Amazon Linux 2 (free tier)**.
4. Select an **Instance Type**, such as **t2.micro (or t3.micro)**.
5. Create or choose an existing **Key Pair** for secure SSH access.
6. Configure **Network Settings** to create a new **Security Group** and allow SSH traffic from your IP address.
7. Click **Launch instance**.
8. If error or warning shows during launch. Just display the error message.
9. If Not error then the instance state changes to **Running**, select the instance, and click **Connect**.

Task 6: Termination or Stop of an EC2 Instance

It's crucial to terminate instances you're no longer using to avoid incurring unnecessary costs.

1. AWS Management Console (GUI)

1. Navigate to the **EC2 service**.
2. Select the instance you want to terminate or stop
3. Click **Instance state** -> **Terminate instance** or **Stop Instance**
4. Confirm the termination or Stop.

Task 7: Answer reflection questions

Answer these following questions:

- What felt different about deploying resources in the cloud vs. using your personal computer?
- How do you think cloud computing changes the way organizations build and deliver software?

Lecturer
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